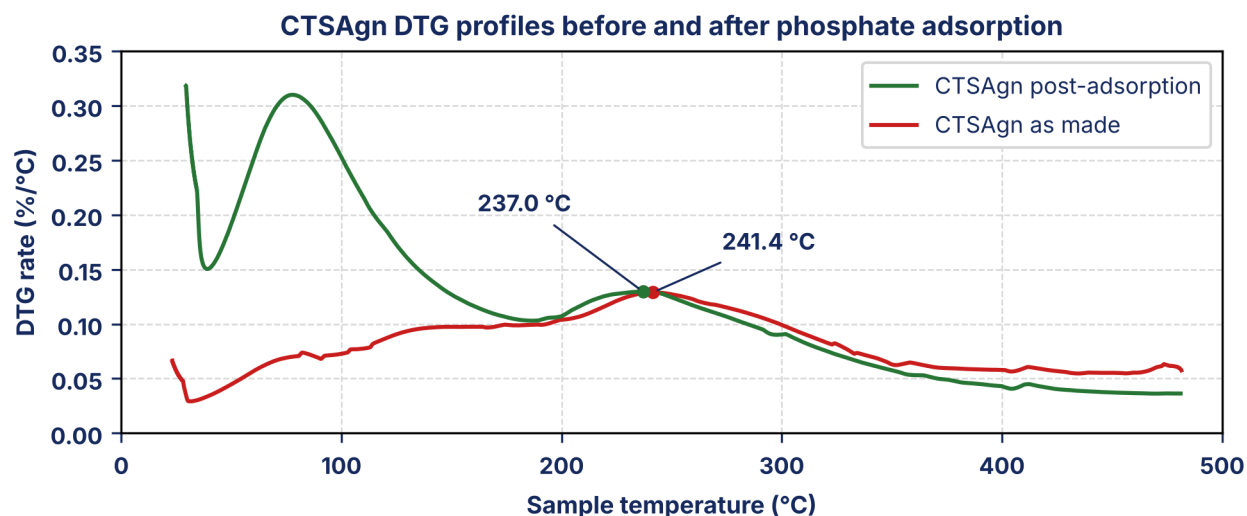


Thermal stability assessment: CTSAgn composite

Primary degradation peaks from the September 2018 TGA campaign, read above the water step. Rates are per cent of the 150 °C mass per °C.

Material	Peak temperature (°C)	Peak DTG rate (%/°C)
Chitosan precursor	305.2	1.120
Alginate precursor	246.8	1.270
CTSAgn as made	241.4	0.129
CTSAgn post-adsorption	237.0	0.130



On the lab's 150 °C basis the as-made composite's peak mass-loss rate is 8.7 times lower than chitosan's and 9.9 times lower than alginate's; after loading the factors are 8.6 and 9.8. The composite's peak rate is essentially unchanged by adsorption (0.129 to 0.130 %/°C, +0.6%). On the instrument's initial-mass column it would appear to fall by 14%, but that is the extra water the loaded sample carries, not a change in how fast the dry material decomposes.

The composite peak at 241.4 °C lies below both precursors, 5.4 °C below alginate and 63.8 °C below chitosan, not between them. Its position and the absence of a chitosan-like event near 305 °C are consistent with an alginate-associated decomposition; the traces alone cannot prove the mechanism.

After loading, the peak moves down 4.4 °C, from 241.4 to 237.0 °C. A lower peak temperature is not by itself evidence of earlier onset. Sorbed phosphate and counter-ions may alter the alginate-rich network, but that is a hypothesis, and with the peak rate unchanged neither observation shows stabilisation or earlier breakdown.

For the recovered material, 5% of the mass held at 150 °C has been lost by about 195 °C under the nitrogen ramp. This is the screening handling/regeneration ceiling, about 42 °C below its peak-rate maximum. It is not a demonstrated safe service limit or a measurement of repeated-cycle capacity retention. [Regeneration_Guidance_Bulletin.pdf](#) quotes a higher blanket figure of 220 °C, drawn from other products and never measured on this lot; the 195 °C from this material's own trace is the better-supported basis.